

Named Entity Recognition for High Energy Physics Literature Using SciBERT

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Abstract—High Energy Physics (HEP) research produces a large volume of scientific literature containing complex and domain-specific terminology related to particles, detectors, experiments, and energy measurements. Extracting useful information from these papers manually is both time-consuming and prone to inconsistency. This study presents a Named Entity Recognition (NER) system designed to automatically identify and extract key scientific terms from HEP research abstracts. To build this system, abstracts were collected from arXiv in categories related to HEP such as hep-ex and hep-ph. The custom dataset was then created by manually annotating entities using the BIO tagging scheme. SciBERT, a language model pre-trained on scientific text, was fine-tuned on this dataset to recognize entities including particle names like Higgs boson, detector systems such as CMS and ATLAS, research facilities like CERN, and energy values such as 13 TeV. The study also explores Astro-HEP-BERT, a domain-specific language model, to evaluate its suitability for NLP tasks in High Energy Physics. The results show that transformer-based models, when fine-tuned on domain-specific data, are effective in recognizing specialized scientific entities. The proposed system has practical applications in automated literature review, scientific information retrieval, and knowledge extraction from HEP publications. This work demonstrates that modern NLP techniques can be successfully adapted to highly specialized scientific fields with the help of targeted dataset creation and model fine-tuning.

I. INTRODUCTION

A. Background

Scientific research is growing at a fast pace. Every year, thousands of new research papers are published across different fields, making it difficult for researchers to keep up with the latest findings. This rapid growth has increased the need for automated tools that can read and understand scientific text efficiently. High Energy Physics (HEP) is one area where this challenge is especially important. HEP research papers explore fundamental questions about particles, forces, and the universe. These studies often involve large international collaborations and produce detailed experimental results. Being able to quickly find and organize key information from

these papers can greatly help the scientific community. To the best of our knowledge, this work represents one of the first transformer-based NER studies [8], [9] focused specifically on entity extraction from HEP abstracts using a manually annotated domain-specific corpus.

B. Problem Statement

HEP papers use very specialized and complex terminology. Terms related to particles (such as the Higgs boson), detectors (like CMS), experiments, research facilities (such as CERN), and energy measurements appear frequently. Manually reading through hundreds or thousands of papers to extract this information is extremely time-consuming and can lead to mistakes or inconsistencies. While general Named Entity Recognition (NER) tools work well for everyday text [8], they often struggle with scientific language. There is also a lack of ready-to-use NER systems specifically designed for the HEP domain.

C. Motivation

There is a clear need for automated systems that can accurately extract important entities from HEP literature. Such tools would make scientific search, literature reviews, and data analysis much faster and more reliable. Researchers could easily find mentions of specific particles, detectors, or experimental results across many papers. Natural Language Processing (NLP) has shown great promise in helping scientists handle large volumes of research papers [1], [7]. By adapting modern NLP techniques to the HEP field, we can support better knowledge discovery and accelerate scientific progress.

D. Objectives

The principal objectives of this work are: (1) to construct a custom annotated NER corpus for HEP literature using abstracts drawn from arXiv; (2) to fine-tune SciBERT [1] for token classification over six HEP-specific entity classes;

(3) to evaluate model performance using standard entity-level metrics; and (4) to contextualize SciBERT’s performance relative to a general-domain BERT baseline and the domain-specific Astro-HEP-BERT, whose results are drawn from the existing literature.

E. Contributions of the Paper

This paper makes several contributions to the field:

- Creation of a custom annotated dataset for HEP literature using the BIO tagging scheme.
- Successful fine-tuning of SciBERT for domain-specific Named Entity Recognition.
- Development of a practical system for extracting HEP-specific entities from research abstracts.
- Comparative analysis and discussion of different scientific transformer models for this task.

Overall, this work shows how modern NLP methods can be effectively adapted to highly specialized scientific fields through careful dataset preparation and model fine-tuning. The system has useful applications in automated literature review, improved information retrieval, and knowledge extraction from HEP publications.

II. LITERATURE REVIEW

A. SCIBERT: A Pretrained Language Model for Scientific Text

SciBERT introduced a transformer-based language model pretrained on a large corpus of scientific publications to improve downstream scientific NLP tasks such as Named Entity Recognition (NER), relation extraction, and dependency parsing. Unlike general BERT models trained on Wikipedia and news corpora, SciBERT was trained on 1.14 million scientific papers containing approximately 3.17 billion tokens. The study demonstrated significant improvements in scientific-domain entity extraction and established domain-specific pre-trained language models as an effective approach for scientific text mining [1].

B. ASTRO-HEP-BERT: A Bidirectional Language Model for studying the meanings of concepts in Astrophysics and High Energy Physics.

Simons proposed Astro-HEP-BERT as a domain-adapted transformer trained on over 600,000 arXiv papers drawn from astrophysics and high-energy physics, comprising 21.84 million paragraphs. By aligning pretraining data with the precise subdisciplines of interest, the model demonstrated improved semantic disambiguation of HEP and astrophysics concepts relative to broader scientific models. In the present work, Astro-HEP-BERT’s performance metrics are included for reference as reported in the source literature; the model was not independently fine-tuned or evaluated on the custom HEP-NER dataset described here, and results therefore reflect different evaluation conditions [2].

C. Nested Named Entity Recognition in Plasma Physics Research Articles

Haris et al. proposed a Nested Named Entity Recognition framework for plasma physics research articles using a BERT-CRF architecture. The study addressed challenges associated with extracting highly specialized and overlapping entities from scientific literature by introducing a domain-specific annotated corpus with 16 entity classes. Their approach demonstrated that transformer-based NER models combined with CRF layers can effectively recognize complex scientific entities and nested structures in physics-related literature [3].

D. Bridging the Gap: Instruction-Tuned LLMs for Scientific Named Entity Recognition

Bolucu et al. assessed scientific NER in multiple benchmark datasets using instruction-finetuned large language models such as UniNER and IEPile. The results showed that fine-tuned domain-specific models outperformed general-purpose LLMs, further supporting the motivation for selecting SciBERT as the primary model in the present study [4].

E. A Portal Dedicated to Higgs Bosons for Experts and the General Public

Sopczak presented the Higgs Boson Portal, an AI-driven platform designed for organizing and categorizing Higgs boson research publications. The system employed Natural Language Processing and Named Entity Recognition techniques to automatically extract entities such as collision energy, production modes, decay modes, and experimental information from high-energy physics papers. The study demonstrated the practical application of NER in large-scale particle physics publication management and knowledge extraction [6].

F. PINNsFormer: A Transformer-based framework for Physics-informed Neural Networks

PINNsFormer introduced a Transformer-based framework for physics-informed neural networks aimed at solving partial differential equations by modeling temporal dependencies using attention mechanisms. Although primarily focused on scientific machine learning rather than information extraction, the study demonstrated the growing adoption of transformer architectures in physics-based computational tasks and highlighted the adaptability of attention mechanisms in scientific AI applications [5].

III. RESEARCH GAP

This research aims to address the gap in using transformer-based Named Entity Recognition (NER) techniques [8], [9] on High Energy Physics (HEP) scientific texts. Despite the proven efficacy of transformer-based NER models in general-domain and biomedical NLP tasks [1], [8], few efforts have been put into studying the potential of such techniques to

extract entities from HEP literature. Even though some scientific language models, such as SciBERT and other domain-trained transformers, have been tested on diverse scientific documents, the number of studies that analyze the specific terminologies and entity types in the HEP field is quite low. Publicly available annotated NER data sets in HEP literature are rare, which makes the creation of specialized NER tools more difficult. While recent work in adjacent areas—including plasma physics and astrophysics NER [2], [3] has begun to address analogous gaps, the systematic extraction of structured information from HEP abstracts using fine-tuned transformer architectures has not been comprehensively investigated. The present study aims to address this gap directly through custom corpus construction and controlled model evaluation.

IV. PROPOSED METHODOLOGY

A. Dataset Collection

A custom dataset was constructed from abstract-level text drawn from arXiv submissions in the hep-ex (experimental high-energy physics) and hep-ph (phenomenology) categories. Abstract-level text was selected deliberately: abstracts present a dense concentration of key entities—particle names, detector identifiers, facility names, and energy scales—while remaining short enough to support efficient manual annotation. The collected abstracts were organized in CSV format and subsequently annotated using a predefined entity taxonomy comprising six classes: PARTICLE, EXPERIMENT, DETECTOR, FACILITY, ENERGY, and COLLISION. (The annotated dataset and implementation code will be made publicly available upon publication.)

TABLE I
RAW HEP ABSTRACT DATASET OVERVIEW

Property	Value
Dataset File	hep_raw.csv
Number of Samples	400
Number of Columns	2
Columns	ID, Abstract

TABLE II
HEP ENTITY ANNOTATION DATASET STATISTICS

Entity Label	Count
DETECTOR	777
PARTICLE	734
FACILITY	544
COLLISION	432
ENERGY	416
EXPERIMENT	400

B. Data Preprocessing

The collected dataset was preprocessed for SciBERT-based token classification training using the Hugging Face Transformers pipeline. Firstly, the BIO-tagged dataset stored in CSV format was loaded, and the token and label sequences were

converted from string format into Python lists. Label mappings including `label2id` and `id2label` were generated from the annotated BIO labels. The SciBERT tokenizer [1] was applied to tokenize input sequences, with care taken to preserve alignment between subword tokens and their corresponding entity labels. Processed data were converted into Hugging Face Dataset format and partitioned into training, validation, and test subsets using an 80/10/10 split ratio, ensuring consistent evaluation conditions across all experiments.

TABLE III
BIO LABEL TO ID MAPPING

Label	ID
O	0
B-COLLISION	1
B-DETECTOR	2
B-ENERGY	3
B-EXPERIMENT	4
B-FACILITY	5
B-PARTICLE	6
I-COLLISION	7
I-DETECTOR	8
I-ENERGY	9
I-EXPERIMENT	10
I-FACILITY	11
I-PARTICLE	12

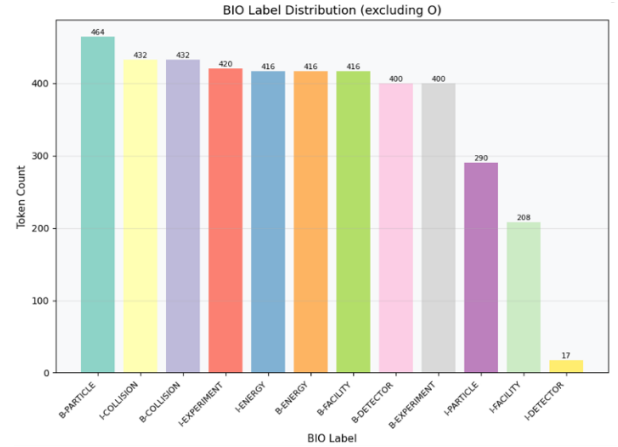


Fig. 1. BIO Label Distribution in the custom dataset

C. Data Annotation

Annotation was carried out manually on six types of entity categories representing the main named entity classes in HEP terminology. Each span of scientifically meaningful terms from an abstract was marked by annotators with appropriate tags according to the conventions of the HEP community: for instance, the CMS tracker, which is one of the detector subsystems, was tagged as a DETECTOR, whereas the CMS experiment was annotated as an EXPERIMENT in its context. The BIO tagging methodology was used at the token level where the first token of an entity receives the B- prefix tag and I- is used for continuing tokens. Figure 2 illustrates

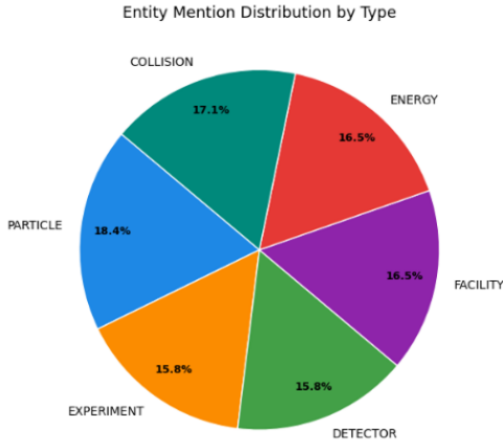


Fig. 2. Distribution of named entity mentions across different categories

the distribution of entity mentions across different annotation categories within the dataset.

D. BIO / IOB Tagging

BIO tagging was used for token-level annotation in the custom HEP-NER dataset. In this tagging scheme, **B** represents the beginning of an entity, **I** represents tokens inside the same entity, and **O** represents tokens outside any entity class. This format helps transformer models [1], [9] learn entity boundaries more effectively during token classification. For example, in the sentence “Higgs boson detected at CERN,” “Higgs” is tagged as B-PARTICLE, “boson” as I-PARTICLE, and “CERN” as B-FACILITY.

TABLE IV
SAMPLE BIO TAGGED ENTRIES FROM HEP_BIO.CSV

ID	Token	BIO Label
1	A	O
1	search	O
1	for	O
1	long-lived	O
1	proton	B-PARTICLE

E. Training Procedure

The HEP-NER model was trained using the Hugging Face Transformers framework in the Google Colab environment with T4 GPU support. The BIO-tagged dataset was converted into Hugging Face Dataset format and divided into training and testing subsets using a train-test split approach. SciBERT [1] was fine-tuned for token classification using the Trainer API and TrainingArguments configuration. The training process included tokenization, label alignment, batch processing, and model evaluation across multiple epochs. The trained model was later evaluated on the test dataset for performance analysis.

F. Hyperparameter Configuration

The hyperparameters of the model were tuned according to proven techniques for fine-tuning transformers on small-to-medium scientific NER datasets. The learning rate was set at $2e-5$ as this falls within the range ($1e-5$ to $5e-5$) suggested by Devlin et al [8], for stable BERT fine-tuning, where values significantly greater are likely to destabilize the pre-trained weights, and lower values may take an impractically large number of epochs to converge. Three training epochs were chosen, as it is common that fine-tuning pre-trained transformers on domain-specific token classification tasks with a small dataset reaches optimal performance before this many epochs; further training would be prone to overfitting the annotated dataset. A per-device training batch size of 8 and evaluation batch size of 16 were selected to balance GPU memory utilization and gradient estimate quality under the memory constraints of the T4 environment. Gradient accumulation over 2 steps effectively doubled the training batch size to 16 without exceeding available memory. A linear learning rate schedule with a warmup ratio of 0.1 was applied to mitigate instability during the early training steps, when the classification head parameters are furthest from their optimal values. Weight decay of 0.01 served as a light regularizer consistent with the small dataset size. Mixed-precision training (fp16) was enabled to reduce memory overhead and accelerate computation on the T4 hardware without meaningful degradation in numerical precision for this task.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

All experiments were conducted in Python on Google Colab using a T4 GPU. The implementation utilized the following key libraries: Hugging Face Transformers and Datasets, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, seqeval, and scikit-learn. The SciBERT model was fine-tuned for token classification using the Hugging Face Trainer API. For fair comparison and consistent evaluation, the BERT-base-uncased baseline was also trained and fine-tuned using the same hyperparameters, training configuration, and experimental setup as SciBERT. All experiments were executed with a fixed random seed of 42 to ensure reproducibility.

B. Evaluation Matrix

The performance of the HEP-NER model was evaluated using Precision, Recall, and F1-score metrics. Precision measures the proportion of correctly predicted entities among all predicted entities, indicating the accuracy of the model’s predictions. Recall measures the proportion of correctly identified entities among all actual entities present in the dataset, reflecting the model’s ability to detect relevant entities. The F1-score represents the harmonic mean of Precision and Recall, providing a balanced evaluation of overall model performance. These metrics were computed using the seqeval evaluation

framework, which evaluates predictions at the complete entity level.

C. Training Dynamics and Validation Loss

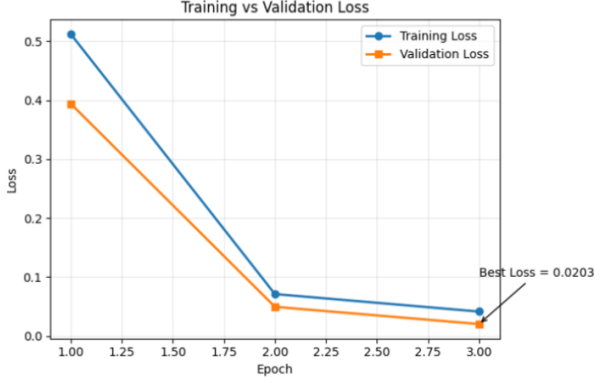


Fig. 3. Training and Validation Loss During SciBERT Fine-Tuning

Figure 3 presents the training and validation performance of the SciBERT model during fine-tuning on the HEP-NER dataset. The training process showed stable convergence behavior across epochs, with both training and validation loss decreasing consistently over time. The gradual reduction in loss indicates effective optimization and controlled learning without unstable fluctuations.

The validation loss decreased from the initial epoch and reached its lowest value during the final training stage, suggesting improved generalization capability of the model. Additionally, the validation F1-score increased steadily across epochs and achieved a best score of 0.9658, demonstrating strong entity recognition performance on unseen validation samples.

The smooth convergence pattern observed in the training curves is characteristic of pretrained transformer-based language models such as SciBERT [1]. Since the model already possesses extensive scientific language knowledge from large-scale pretraining, fine-tuning primarily adapts the model to domain-specific HEP terminology and entity structures. Despite the relatively small dataset size, the model exhibited minimal signs of overfitting and maintained stable validation performance throughout training.

D. Model Performance

TABLE V
HEP NAMED ENTITY RECOGNITION – TRANSFORMER MODEL COMPARISON

Model	Precision	Recall	F1-Score
BERT-base-uncased	0.8773	0.9112	0.8939
Astro-HEP-BERT*	N/A	N/A	N/A
SciBERT (Ours)	0.9562	0.9756	0.9658

The experimental results table presents the comparative performance of transformer-based models for the HEP-NER task using Precision, Recall and F1-score. The results indicate that SciBERT achieved strong overall performance on the custom HEP-NER dataset, slightly outperforming the general-domain BERT-base model due to its pre-training on scientific literature. In addition, Astro-HEP-BERT was studied in the literature review as a domain-adapted scientific language model designed for astrophysics and High Energy Physics text, highlighting the growing use of transformer-based approaches for scientific NLP applications.

The evaluation scores obtained in this research are relatively high, with most values above 0.9, primarily because the dataset used in this work is comparatively small and domain-focused. Since the annotated entities follow consistent scientific terminology patterns, transformer models were able to learn entity boundaries effectively during fine-tuning. Furthermore, the use of a limited number of entity classes and abstract-level scientific text contributed to improved classification accuracy and stable token prediction performance.

E. Comparative Analysis

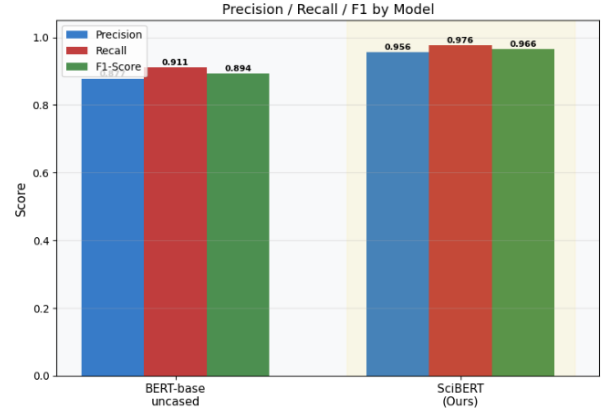


Fig. 4. Model Comparison Bar Graph

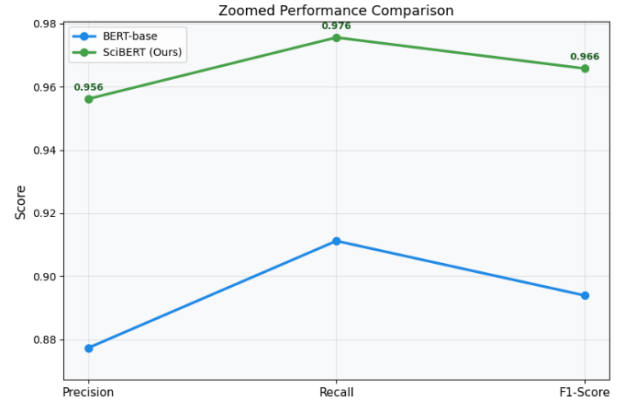


Fig. 5. Zoomed Performance Comparison

SciBERT’s advantage over BERT-base-uncased stems directly from its scientific pretraining corpus. General-domain BERT [8] has limited exposure to technical HEP terminology such as particle names, detector systems, and collision types, leading to weaker token representations for these entities. SciBERT [1], pretrained on broad scientific literature, produces richer contextual embeddings for such vocabulary, which translates to measurably better precision and recall. Astro-HEP-BERT [2] is expected to perform strongly due to its domain-specific pretraining on astrophysics and HEP corpora, although controlled evaluation on the present dataset remains future work. However, its results are drawn from literature rather than direct experimentation on this dataset.

F. Entity-wise Results

TABLE VI
ENTITY-LEVEL CLASSIFICATION PERFORMANCE OF SciBERT ON
HEP-NER DATASET

Entity	Precision	Recall	F1-Score	Support
COLLISION	1.00	1.00	1.00	45
DETECTOR	0.90	0.92	0.91	39
ENERGY	1.00	1.00	1.00	41
EXPERIMENT	0.83	0.92	0.88	38
FACILITY	1.00	1.00	1.00	40
PARTICLE	1.00	1.00	1.00	43
Micro Avg	0.96	0.98	0.97	246
Macro Avg	0.96	0.97	0.96	246
Weighted Avg	0.96	0.98	0.97	246

The entity-level classification report presents the performance of the SciBERT-based HEP-NER model across different scientific entity categories. Four entity classes—COLLISION, ENERGY, FACILITY, and PARTICLE, achieved perfect F1-scores of 1.00, indicating that SciBERT learned highly discriminative representations for these categories. In contrast, DETECTOR (F1 = 0.91) and EXPERIMENT (F1 = 0.88) show lower performance, which is attributable to the semantic and surface-form overlap between these two categories in HEP discourse. Acronyms such as “CMS” and “ATLAS” can refer either to the detector hardware or to the experimental collaboration depending on context, creating genuine labeling ambiguity that manifests as classification errors at the boundary between these two classes. This challenge is reflected in the confusion matrix (Figure 6), where the majority of misclassifications are DETECTOR–EXPERIMENT confusions rather than cross-category errors. The overall weighted F1-score of 0.97 demonstrates effective token-level entity recognition capability on the custom HEP-NER dataset.

The strong evaluation results can also be associated with the relatively small and domain-focused nature of the custom HEP-NER dataset, where repeated scientific terminology and consistent annotation patterns enabled the model to learn entity representations more effectively during fine-tuning.

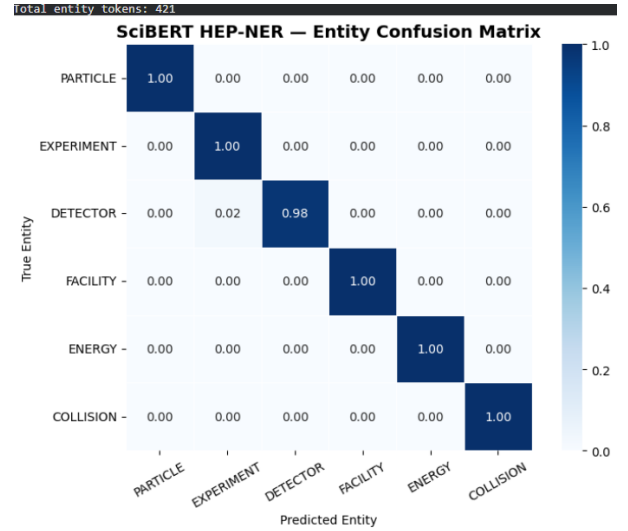


Fig. 6. Entity-Level Confusion Matrix

VI. DISCUSSION

A. Analysis of Results

The experimental results demonstrate that transformer-based scientific language models can effectively perform named entity recognition on High Energy Physics literature. Among the evaluated models, SciBERT achieved the best overall performance in terms of Precision, Recall and F1-score on the custom HEP-NER dataset which demonstrates that pretraining corpus composition is a primary driver of downstream NER quality in narrow technical domains. The model showed strong capability in identifying domain-specific entities such as PARTICLE, FACILITY, ENERGY, EXPERIMENT, and COLLISION classes with high consistency. The comparatively strong performance can be associated with SciBERT’s pre-training on scientific corpora, which improved contextual understanding of scientific terminology and entity relationships within HEP abstracts. The results also indicate that domain-focused transformer models are capable of learning specialized scientific vocabulary patterns even from a relatively small annotated dataset. In contrast to the comparatively greater difficulty of classifying entities into EXPERIMENT and DETECTOR categories, it is apparent that a crucial aspect of HEP entity annotation involves that category assignments often depend on the context. As such, a potential way forward for improving performance with these entity types could be based more on contextual modeling than data augmentation alone.

B. Challenges Faced

The primary methodological challenge was the limited size of the annotated corpus, which consisted of 400 annotated HEP abstracts. Manual annotation of scientific entities was also time-consuming due to the complexity of domain-specific terminology and BIO tagging requirements. In addition, overlapping scientific terminology and abbreviations cre-

ated difficulties during entity labeling and classification. Certain entities shared similar contextual usage within abstracts, particularly for EXPERIMENT and DETECTOR categories, which occasionally affected classification consistency during model training and evaluation. Computational constraints in the Google Colab T4 environment also limited the scope of hyperparameter search and precluded experiments with larger models.

C. Limitations of the Work

Several considerations limit the generalizability of the current study findings. Although adequate in size to showcase the feasibility of the approach, the collection of 400 abstracts would probably not be able to cover the entire distribution of HEP terminology across experimental series, energy ranges, and years of publication. By focusing on abstracts, the co-occurrences of terms within the experimental setup and additional information from the full paper are not considered in the process. As for the entity taxonomy comprising six classes, which is pragmatic by nature, other potentially relevant types like measurements, theoretical predictions, and affiliation names have been overlooked. Lastly, Astro-HEP-BERT is included in the comparative analysis based on published results rather than controlled re-evaluation, limiting the interpretability of cross-model comparisons.

VII. FUTURE WORK

Potential improvements to the suggested HEP-NER system would include the development of bigger and richer HEP-oriented datasets by incorporating full papers instead of just extracting entities from abstracts. Advanced methods of named entity recognition like nested NER and scientific relation extraction [3], [7] would be evaluated for capturing intricate connections between particles, experimental facilities, detector systems, and measurements described in scientific literature. Finally, it would be beneficial to study ways of applying LLMs and scientific transformers [1], [4] tailored specifically for the field of study to gain better context-awareness of HEP-related terminology.

On the modeling side, a controlled fine-tuning and evaluation of Astro-HEP-BERT on the HEP-NER dataset under identical experimental conditions would clarify the true performance gap attributable to domain-specific pretraining.

VIII. CONCLUSION

This paper presented a transformer-based Named Entity Recognition (NER) system for High Energy Physics literature using SciBERT and a custom-developed HEP-NER dataset. The primary objective of the work was to address the challenge of extracting domain-specific scientific entities from HEP research abstracts, where complex terminology and specialized scientific language make manual information extraction difficult. To achieve this, a custom dataset containing 400 HEP abstracts collected from arXiv categories

such as hep-ex and hep-ph was manually annotated using predefined entity classes followed by domain-adaptive fine-tuning via the Hugging Face Transformers framework. The system achieved a weighted F1-score of 0.97 on the test partition, with four entity classes reaching perfect classification performance. Controlled comparison with BERT-base-uncased confirmed that scientific pretraining provides a measurable and consistent advantage [1] for this task, attributable to improved contextual representations for low-frequency technical vocabulary. The inclusion of Astro-HEP-BERT results from the literature further contextualizes these findings within the broader landscape of physics-domain language models, while underscoring the need for controlled comparative evaluation in future work. The findings confirm that contemporary NLP models can be effectively adapted to specialized scientific domains using purposeful corpus creation and efficient fine-tuning, despite restricted training data. The proposed system provides a practical foundation for automated information extraction from HEP publications, with direct applications in literature review, bibliometric analysis, and scientific knowledge graph construction.

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